

# Standardization Document -

INST490 iConsultancy—SPLC Voting & Civic Participation Project  
Spring 2026 Section 103 (Alabama Black Belt)

## About This Document

This document contains standardized conventions that are followed across the whole BB-DIG project. It also contains the scope of our project as well as any known limitations in our project.

## 1. File/Folder Names

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### 1.1 Dataset Folders

**Naming format:** datasettype\_year\_electiontype

- **Example:** voter\_turnout\_2024\_general

### 1.2 Datasets

**Naming format:** datasettype\_year\_electiontype.csv

- **Example:** voter\_turnout\_2024\_general.csv
- **Note:** general means general election

### 1.3 Data Dictionary

**Naming format:** datasetname\_data\_dictionary.xlsx

- **Example:** voter\_turnout\_2024\_general\_data\_dictionary.xlsx

### 1.4 Data Documentation

**Naming format:** datasetname\_documentation.md

- **Example:** voter\_turnout\_2024\_general\_documentation.md

### 1.5 Script Folders

**Naming format:** datasetname\_purpose

- **Example:** voter\_registration\_general\_2020\_cleaning

### 1.6 Scripts

**Naming format:** datasetname\_purpose

- **Example:** voter\_registration\_general\_2020\_cleaning.ipynb

## 1.7 Scripts

**Naming format:** datasetname\_purpose

- **Example:** voter\_registration\_general\_2020\_cleaning\_documentation.md

## 1.8 Deliverable Documentation

**Naming format:** deliverable\_lastdateofrevision

- **Example:** standardization\_0517

# 2. Data Structure Rules

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## 2.1 General Rules

- Every final CSV must have one row per county
- No total or summary rows in the final output, only county specific rows
- All percentage fields stored as decimals, for example 0.65 not 65%
- Missing values documented as “null” rather than filled 0 or blank

## 2.2 Geographic Identifiers

- Every final CSV must have both county\_name and fips\_code as the first two columns
- county\_name must be title case with no extra spaces or line breaks
- FIPS Code:
  - The FIPS codes for all of the Alabama counties are listed here: <https://transition.fcc.gov/oet/info/maps/census/fips/fips.txt>
  - For our fips\_code we stored them as a 4 digit integer as provided by the Redistricting Hub. Standard practice in the documentation above is 5 digit codes with leading zeroes such as 01005 instead of just 1005. We keep them as used in RDH for data comparison, but future use could pad with leading zeros if needed.
  - FIPS codes are joined into every dataset from RDH using county\_name as the merge key

## 2.3 Column Names

- All column names must be lowercased with underscores separating individual words
  - **Example:** a column called Registered Voters should be called registered\_voters
- No spaces, no camelCase, no abbreviations unless documented
- The column name format will adhere to the following format:  
descriptor\_demographiccategory
  - **Example:** turnout\_rate\_black, active\_registered\_white, eligible\_hispanic

## 3. Scopes

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### 3.1 The 17 Black Belt Counties

The project covers 17 Alabama Black Belt counties: Barbour, Bullock, Butler, Choctaw, Crenshaw, Dallas, Greene, Hale, Lowndes, Macon, Marengo, Montgomery, Perry, Pike, Russell, Sumter, and Wilcox.

Every script we have filters to this exact list. Counties outside of this list are excluded from any final outputs.

### 3.2 Race Scope Decisions

- Final outputs only focus on Black, White, and Latinx race columns, aligned with the SPLC focus area
- Asian, American Indian, Other, Not Identified, and any political identifiers are excluded from race-specific columns
- Their counts however still may appear in totals where it is applicable such as the `active_total`, `inactive_total`, and `total_ballots_cast`
- Hispanic data from source data is renamed to Latinx in all final outputs to align with SPLC language

## 4. The Three Dataset Types

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- This project produces three types of datasets across 2020 and 2024 (six CSVs total):
  - `eligible_population` files measure participation using CVAP eligible population as the denominator. Answers: of everyone who could have voted, how many did?
  - `voter_turnout` files measure turnout using ALVR total registered voters (active plus inactive) as the denominator. Answers: of registered voters, how many showed up?
  - `voter_registration` files measure registration access and purge risk using ALVR-only data. Answers: who registered and who is at risk of being removed from the rolls?
- Each dataset uses a different denominator because each answers a different question. They should be used together for the complete picture of civic participation.

## 5. Source Hierarchy

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- **Alabama Secretary of State (SoS)** is the primary source for voter turnout and registration
- **ALVR (Alabama Voter Registration October sheet)** is the primary source for active and inactive registration counties by race
- **CVAP (Census Bureau)** is the primary source for eligible population by race

- **RDH (Redistricting Hub)** is used for verification and FIPS codes only; not included in final calculated columns
- Any discrepancies between the sources must be documented in the dataset documentation.

## 6. Methodology Decisions

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- For voter\_turnout race rates, we use ALVR total registered (active plus inactive) as the denominator because it matches how the overall turnout\_rate is calculated and avoids the CVAP undercount issues
- For voter\_registration, we calculate active\_share (active divided by total registered) using ALVR data only because using CVAP as the denominator produced rates above 100% in most Black Belt counties due to known CVAP undercount in rural majority-minority communities
- For eligible\_population, missing voters are calculated as eligible minus voted, with negative results capped at 0 to handle source disagreement in small populations

## 7. Dataset Folders Format

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Every cleaned CSV must have two corresponding documentation files:

### Part 1: Column Reference (Excel)

- Column name, description, source, type, and an example value
- One row per column in the dataset
- Stored as a .xlsx file

### Part 2: Descriptive Documentation (Markdown)

- Overview with cross-reference paragraph explaining how the file relates to the other datasets
- Sources with version notes
- Limitations specific to that dataset
- Notes including information regarding our methodology
- Intended use

## 8. Script Folders Format

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Every script must have the following corresponding documentation file:

### Descriptive Documentation (Markdown)

- Overview of what the script does, as well as its purpose.
- The sources from which the datasets used in the script are from.

- A “How to Use” section which details any terminology users should be familiar with in order to understand the script outputs.

## 9. Steps for Standardization

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- County names: strip whitespace, remove line breaks, convert to title case across all files
- Add FIPS codes by merging in from RDH
- Rename all columns to follow consistent naming convention
- Strip line breaks from 2020 race file column headers since they all have \n in them
- Filter all files down to the 17 Black Belt counties
- ALVR has a broken two-row header which must be handled by manually setting column names and dropping the first two rows
- Force float conversion using `astype(float)` before division operations to ensure rounding works correctly
- Round all rates to 4 decimal places; drop trailing zeros
- Apply `fillna("null")` before exporting to CSV

## 10. Known Limitations Across the Project

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- CVAP undercount in rural majority-minority counties leads to source disagreement
- Small population counties produce unstable rates especially for Latinx voters
- Butler County has a 418 ballot discrepancy between SoS gender/age files and the SOS total file
- Small SoS internal discrepancies between breakdown files and the total file across most counties
- ALVR October snapshot may differ slightly from Election Day registration totals